

MRI and Electromagnetic Imaging Super-Resolution post-processing methods: Error Feedback Guided Refinement

Abstract

MRI plays a critical role in clinical applications like stroke or brain tumor diagnosis, but its spatial resolution is limited by hardware, acquisition time, motion, and SNR constraints. These issues are amplified in low-field MRI and electromagnetic imaging (EMI) systems [1], leading to inherently low-resolution images. Consequently, images often exhibit blurred boundaries, partial volume effects, and reduced visibility of small lesions. Numerous super-resolution techniques have been proposed to address these limitations, including both classical and deep learning-based methods. Among them, EDSR [2] and ISTA-Net+ [3] are considered the most clinically reliable, while GAN-, diffusion-, and transformer-based approaches face challenges related to realism, complexity, and clinical validation. Super-resolution reconstructs high-resolution anatomical detail from low-resolution data, with deep learning-based convolutional networks leading this field due to their ability to capture complex spatial patterns. In MRI and electromagnetic imaging, SR methods aim to both increase spatial resolution and improve image quality. EDSR performs explicit upsampling by recovering lost high-frequency details, allowing clearer depiction of cortical anatomy, tumor margins, ventricles, and tissue boundaries. In contrast, ISTA-Net+ enhances image quality without altering spatial resolution by reducing noise and improving contrast through an optimization-inspired deep unfolding strategy. Because many medical imaging applications, particularly EM imaging, require true spatial upscaling, this study focuses on EDSR and introduces an Error Feedback Guided Refinement (EFGR) approach to further improve reconstruction accuracy and resolution.

Introduction

EDSR includes several adjustable parameters, but these are mainly network architecture and training hyperparameters rather than image-specific settings. Such parameters include the number of residual blocks, feature channels, upscaling factors, learning rate, optimizer configuration, loss function, and training data setup. Unlike model-based approaches, these settings determine the network's learning capacity and behavior and remain fixed during inference, which is non-iterative, deterministic, and stable.

Improper hyperparameter selection in EDSR typically results in reduced performance rather than system failure. For instance, an insufficient number of residual blocks or channels may lead to underfitting, whereas excessive depth can cause overfitting or increased memory demands. Inappropriate learning rates can slow convergence or destabilize training, and discrepancies between training and testing data can limit generalization. Nevertheless, even sub-optimally tuned models usually generate visually plausible, artifact-free outputs, with degradation mainly observed in sharpness and PSNR.

Performance degradation from parameter misconfiguration in EDSR is generally moderate. Minor tuning errors, such as slightly incorrect network depth or learning rate, often reduce PSNR by 0.1–0.5 dB, while more noticeable misadjustments can cause losses of 0.5–1.0 dB. Larger PSNR drops, exceeding 1 dB, are usually associated with significant preprocessing errors or domain shifts rather than typical hyperparameter choices. Severe degradation of several dB occurs only in extreme cases, such as inadequate training or evaluation on entirely different imaging domains.

When EDSR is applied to extremely low-resolution data, such as MRI or electromagnetic images with slice thicknesses of 3–5 mm, substantial PSNR reductions may occur even with well-trained models. This limitation arises because critical high-frequency anatomical information is absent, making the super-resolution task inherently ill-posed. Consequently, the network cannot reliably reconstruct missing details, often producing blurred or oversmoothed results. These issues stem from information loss and domain mismatch, not from instability in the EDSR framework.

For very low-resolution or out-of-domain inputs, super-resolution performance is fundamentally constrained, and results must be interpreted cautiously to avoid misleading anatomical conclusions. Advanced strategies such as domain adaptation, iterative refinement, or GAN-based enhancement may offer partial improvements. In

this project, two Error Feedback Guided Refinement (EFGR) methods are proposed to further improve resolution and reconstruction accuracy for low-resolution imaging data. Due to limited availability of EMI data, the proposed methodology and evaluation are conducted using MRI data.

Algorithm Overview

The EFGR framework employs a set of spatial-domain and frequency-domain post-processing strategies to enhance the outputs of super-resolution (SR) networks. Its primary objective is to recover fine structural details that may not be fully captured by the base SR model. This is achieved through operations such as frequency-band enhancement, edge-aware filtering, high-frequency compensation, spatial smoothing guided by anatomical priors, and artifact suppression. Acting as a secondary reconstruction stage, EFGR improves quantitative metrics such as PSNR and SSIM, while also enhancing perceptual image quality.

In this study, an implementation of EDSR network [4] is trained on a dataset of 423 images from [BrainTumorMRI dataset](#) to generate high-resolution reconstructions from low-resolution inputs. EDSR serves as the primary super-resolution backbone. To further improve reconstruction quality, we introduce post-processing refinement framework termed Frequency Error Feedback Guided Refinement (F-EFGR) and spatial Error Feedback Guided Refinement (S-EFGR). F-EFGR and S-EFGR apply pixel-wise corrections to the EDSR output by propagating estimated reconstruction errors through adaptive weighting functions derived from spatial edge information and frequency-domain energy measures. The present report focuses specifically on the spatial-only (S-EFGR) and frequency-only (F-EFGR) refinement pathways.

Network Architecture

The EDSR model is a high-capacity residual convolutional neural network consisting of three main components:

- **Head module:** initial convolutional layers for feature extraction
- **Body module:** a sequence of residual blocks without batch normalization
- **Tail module:** pixel-shuffle layers for upsampling

Each residual block contains two 3×3 convolutional layers followed by ReLU activation and a residual skip connection. Batch normalization is intentionally omitted to preserve intensity scaling, which is critical for maintaining quantitative consistency in medical imaging applications.

Once a super-resolved image is produced by EDSR (or any SR model that increases spatial resolution), a refinement stage is applied to correct residual reconstruction errors using low-resolution (LR) consistency analysis.

The super-resolved image is first downsampled to the LR space, and the LR reconstruction error is computed as:

$$e_{LR} = SR_{down} - LR$$

This error is then up-sampled back to high-resolution space:

$$e_{up} = \text{Resize}(e_{LR})$$

The resulting high-resolution error map is spatially aligned with the EDSR output and serves as the basis for adaptive refinement.

Spatial Error Feedback Guided Refinement (S-EFGR): Spatial refinement relies on local gradient magnitude, computed as:

$$G = \sqrt{G_x^2 + G_y^2}$$

A normalized gradient map controls the correction strength:

$$\alpha_s = \alpha_{max} - (\alpha_{max} - \alpha_{min}) \cdot \frac{G}{\max(G)}$$

where $\alpha_{max} = 0.25$ and $\alpha_{min} = 0.05$.

The refined image is obtained as:

$$I_{S-EFGR} = SR - \alpha_s \cdot e_{up}$$

This formulation applies to weaker corrections near edges while enforcing stronger corrections in smooth regions, thereby preserving anatomical boundaries.

Frequency Error Feedback Guided Refinement (F-EFGR): Frequency-based refinement estimates local high-frequency content using a Laplacian operator:

$$L = | \nabla^2 SR |$$

The normalized frequency weighting is defined as:

$$\beta_f = \beta_{max} - (\beta_{max} - \beta_{min}) \cdot \frac{L}{\max(L)}$$

with $\beta_{max} = 1.0$ and $\beta_{min} = 0.6$.

The refined output is computed as:

$$I_{F-EFGR} = SR - \beta_f \cdot e_{up}$$

This approach suppresses over-correction in high-frequency regions such as edges, while compensating for low-frequency reconstruction bias.

Evaluation Metrics: Quantitative performance is assessed using the following metrics:

- **Peak Signal-to-Noise Ratio (PSNR):**

$$PSNR = 10 \cdot \log_{10} \left(\frac{MAX^2}{MSE} \right)$$

- **Structural Similarity Index (SSIM):** measures structural fidelity between reconstructed and reference images.

Visualization and Analysis

Figure 1 presents the training results of the EDSR model. In this work, we train an Enhanced Deep Super-Resolution (EDSR) network on a dataset of 423 MRI images from the [BrainTumorMRI](#) dataset, primarily using T1 weighted scans and generating high-resolution reconstructions from their low-resolution counterparts. The EDSR network is used as the primary super-resolution backbone.

Low-resolution and high-resolution image pairs were generated using Lanczos3 down-sampling, with additional down-sampling strategies such as bicubic interpolation planned for future experiments. The model was optimized using an L1 loss function and the Adam optimizer, with a fixed upscaling factor of $\times 2$. Training was conducted in a patch-based manner using $48 \times 48 \times 1$ patches, enabling efficient memory usage and improved learning of fine anatomical details.

Representative visual results, along with the corresponding spatial (α) and frequency (β) weighting maps, are shown in Figures 2, 3, and 4. The associated quantitative results are summarized in Table I.

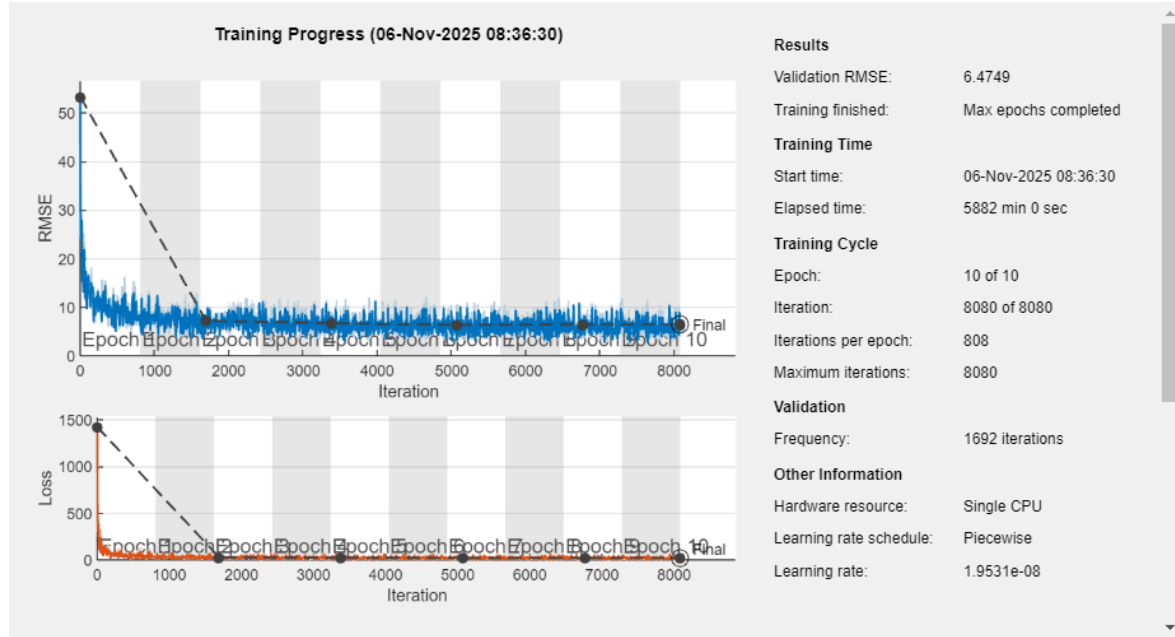
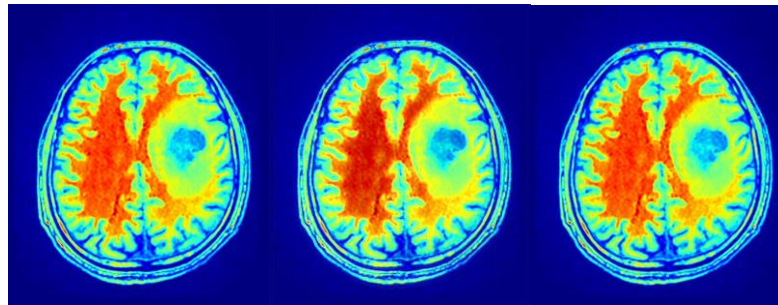


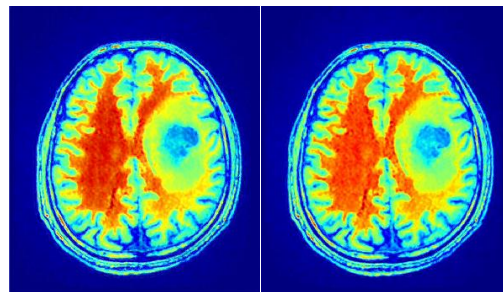
Fig. 1. Training results of EDSR network.



Original image

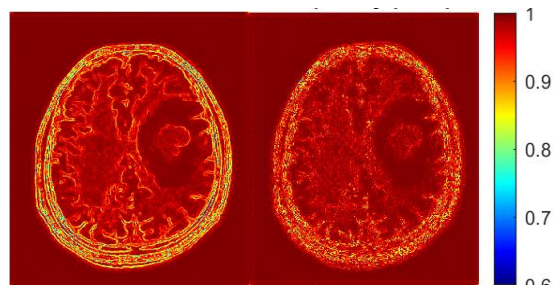
EDSR output

LR up-scaled



S-EFGR

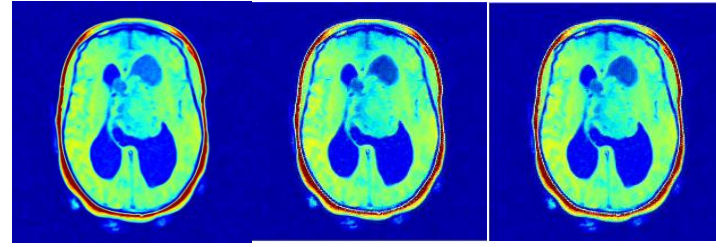
F-EFGR



spatial α map

frequency β map

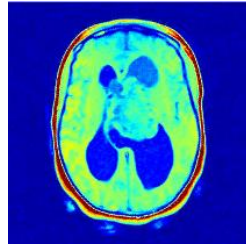
Fig. 2. Result of post processing on glioma



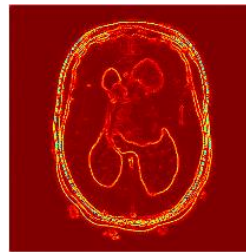
neurocytoma

EDSR output

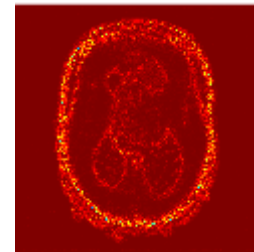
S-FEGR



F_FEGR

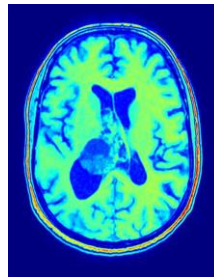


spatial α map

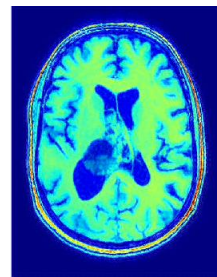


frequency β map

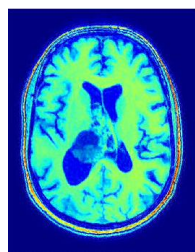
Fig. 3. Result of post processing on neurocytoma



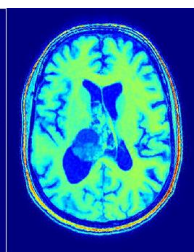
neurocitoma



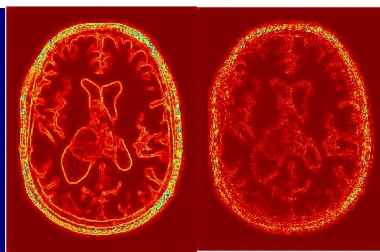
EDSR output



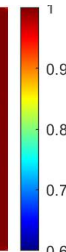
S-FEGR



F-FEGR



spatial α map



frequency β map

Fig. 4. Result of FEGR on neurocytoma

Table I. quantitative results related with figures 2, 3, and 4.

Figure	Method	PSNR (dB)	SSIM
2	EDSR	23.0813	0.9812
2	EDSR+S-FEGR	24.5214	0.9861
2	EDSR+F-FEGR	29.8402	0.9954
3	EDSR	24.2917	0.9837
3	EDSR+S-FEGR	25.6064	0.9874
3	EDSR+F-FEGR	30.6490	0.9955
4	EDSR	20.9992	0.9660
4	EDSR+S-FEGR	22.0291	0.9724
4	EDSR+F-FEGR	25.3747	0.9867

Discussion

The FEGR framework is effective because it is grounded in the physics of image acquisition and the statistical properties of MRI data. Its downsample–error–upsample feedback loop provides a reliable estimate of reconstruction errors without requiring ground-truth images, ensuring consistency with the original low-resolution acquisition process. Spatial refinement selectively applies stronger corrections in smooth anatomical regions while protecting high-gradient areas, improving low-frequency consistency without distorting critical tissue boundaries. Frequency-based refinement further exploits the distinct behavior of low- and high-frequency components in medical images, aggressively correcting low-frequency bias while limiting correction in high-frequency regions to preserve fine anatomical details—yielding substantial quantitative improvements, including PSNR gains of up to 7 dB over baseline EDSR results. Importantly, FEGR functions as a model-agnostic post-processing framework: although demonstrated using EDSR, it can be applied to the outputs of any low-to-high resolution reconstruction or super-resolution algorithm. Moreover, because electromagnetic (EM) images share characteristics with very low-resolution MRI—such as smooth regions, blurred boundaries, and frequency-dependent reconstruction errors—the FEGR refinement strategy can be naturally extended to EM imaging applications. Overall, FEGR complements existing reconstruction models rather than replacing them, correcting residual low-frequency and spatially coherent errors while preserving edges, structural consistency, and both perceptual and numerical image quality, making it a robust enhancement strategy for diagnostic brain MRI and electromagnetic imaging workflows.

References

- [1] B. Lim, S. Son, H. Kim, S. Nah and K. M. Lee, "Enhanced Deep Residual Networks for Single Image Super-Resolution," 2017 IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), Honolulu, HI, USA, 2017, pp. 1132-1140, doi: 10.1109/CVPRW.2017.151.
- [2] Lin, Kehan. "The performance of single-image super-resolution algorithm: EDSR." 2022 IEEE 5th international conference on information systems and computer aided education (ICISCAE). IEEE, 2022.
- [3] Zhang, Jian, and Bernard Ghanem. "ISTA-Net: Interpretable optimization-inspired deep network for image compressive sensing." Proceedings of the IEEE conference on computer vision and pattern recognition. 2018.
- [4] <https://de.mathworks.com/matlabcentral/fileexchange/87799-edsr-enhanced-deep-super-resolution-matlab-port>